

Temperature-controlled cell lysis of modified oleaginous *Saccharomyces cerevisiae* using RNA thermosensors

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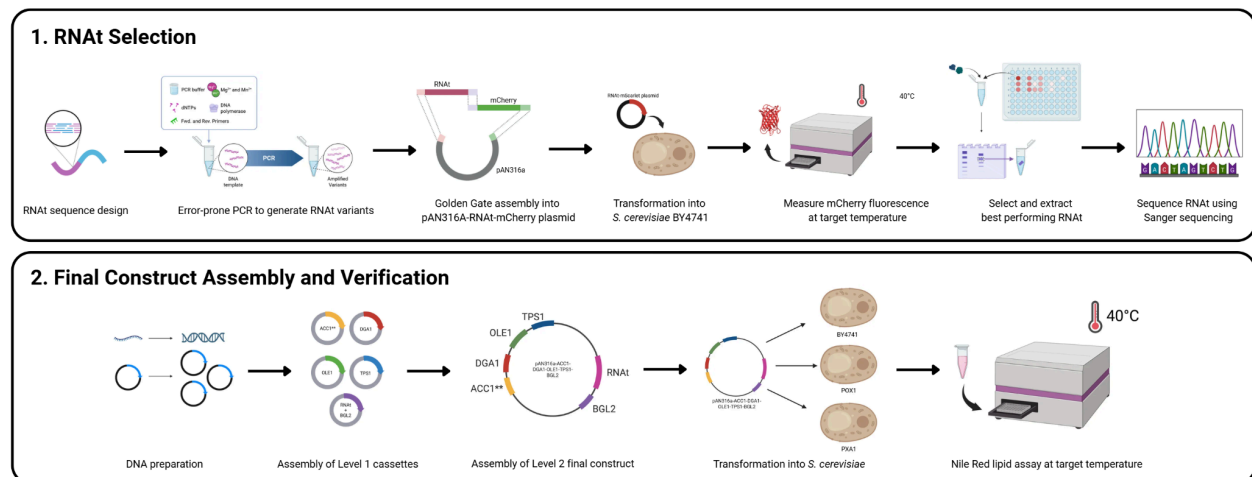
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Abstract

Our project aimed to engineer *Saccharomyces cerevisiae* to overexpress lipids coupled with lysis driven by temperature control. Through computational modeling, a pool of a variety of RNAs narrowed down to the three most optimal. These sequences successfully yielded melting temperatures between 36-44°C. Lipid accumulation was further validated through Nile Red fluorescence readings comparing wild-type BY4741 yeast strain and BY4741 Δ PXA1 yeast strain, which showed higher levels of lipids in the knock out strain than wild type. PXA1 is a gene that brings lipids to the peroxisomal membrane for their degradation; by deleting it, more lipids built up. The other strategy to overexpress lipid synthesis genes such as ACC1** and DGA1 were not able to be validated experimentally due to optimization constraints. Thus, while the theory of the project was verified through modeling data and literature, more troubleshooting for *in vitro* experiments should be conducted in the future.



Introduction

Acetyl-coenzyme A carboxylase 1, encoded by gene ACC1, catalyzes the rate-limiting step of *de novo* fatty acid synthesis, where it is primarily involved in the carboxylation of acetyl-CoA to malonyl-CoA within the lipid synthesis pathway of *S. cerevisiae* (Ferreira et. al, 2018). However, it is also heavily regulated natively, thus a double-mutant version of acetyl-CoA carboxylase (ACC1**) that bypasses its normal metabolic repression will be used. The DGA1 gene encodes Diacylglycerol acyltransferase, which catalyzes the terminal step of fatty acid synthesis involving the storage of triacylglycerol (TAG) in droplet form (Saccharomyces Genome Database [SGD], n.d). Diacylglycerol acyltransferase operates by acylating diacylglycerol (DAG) using acyl-CoA as the donor. Acyl-CoA desaturase 1 (OLE1) works to desaturate saturated fatty acyl-CoA substrates by introducing a cis double bond at the $\Delta 9$ position (Uniprot, n.d). TPS1 encodes trehalose-6-phosphate synthase 1, which synthesizes trehalose, a storage

carbohydrate typically induced by stress conditions such as desiccation, heat, and oxidative stress (SGD, n.d). Trehalose is a very stable non-reducing sugar, with initial melting temperatures as high as 97°C, and resistance to acid-induced hydrolysis (Richards et. al, 2002).

Endo-beta-1,3-glucanase, encoded by BGL2, is involved in cell wall organization, but its overexpression leads to increased glucan degradation within the cell wall (SGD, n.d).

RNA thermometers (RNA_t) are regions in the 5' untranslated region (5'UTR) of mRNA that form secondary structures, acting as thermosensors to regulate translation. Initiation of translation relies on the accessibility of the ribosome binding site (RBS), which is located within the 5'UTR (Sharma et. al, 2022). At low temperatures, the RNA_t adopts a closed conformation that obstructs access to the RBS, which melts at high temperatures to allow translation of the downstream coding sequence. While RNA_t naturally exists in bacteria, Neupert et al. (2009) developed a synthetic RNA_t containing a simpler single-stem loop design as a tool for temperature-controlled expression, which we adopted in the 5'UTR of BGL2 to allow for its overexpression for cell lysis at specific temperatures.

Chassis selection is a critical technical decision, as the characteristics of the host organism influence the feasibility and outcome of experiments and strategies used to construct the final product. Given the objective to engineer a microorganism capable of lipid overproduction for cultivated meat applications, we evaluated potential chassis organisms based on food safety, characterization, availability of genetic engineering tools, and narrowed our candidates to two yeast species, *Yarrowia lipolytica* (*Y. lipolytica*) and *Saccharomyces cerevisiae* (*S. cerevisiae*).

Both species are well-established in the food and biotechnology industries and are generally regarded as safe (GRAS) (Wang et al., 2020), satisfying our criterion for food safety. *Y. lipolytica* is a naturally oleaginous yeast with a high capacity for lipid accumulation, making it a strong candidate for lipid production (Wang et al., 2020). However, engineering flexibility was prioritized over lipid production alone. Thus, *S. cerevisiae* was selected due to its fully characterized genome, extensive available data, rapid growth, and expansive biology toolkit (Fakas, 2016). Together, these characteristics make *S. cerevisiae* a robust and flexible chassis for engineering lipid production.

Methods

Research Approach

The project began with research and design of RNA_t sequence and structure, followed by *in-silico* validation and testing. Relevant genes and parts were obtained, including, the pAN316a shuttle vector from the Nguyen Lab at the University of Toronto; the OYC-dropout-suGFP dropout plasmid and OYC parts, promoters, and terminators from the 2024 and 2025 iGEM

distribution kits; ACC1**, DGA1, OLE1, TPS1, BLG2 from Twist Bioscience; RNAt, mScarlet, and all relevant primers from Genscript. *E. coli* DH5 α was obtained from the Hoare Lab at McMaster University, and *S. cerevisiae* BY4741, *S. cerevisiae* POX1, *S. cerevisiae* PXA1 were obtained from the Corey Lab at the University of British Columbia. Single stranded DNA templates were used to synthesize double stranded DNA using PCR, and plasmids were transformed into *E. coli* DH5 α to clone and were extracted using plasmid miniprep. RNAt mutants were to be generated using Error-Prone PCR and tested alongside the non-mutagenized sequences to determine the sequence with the fastest conformational change and highest expression beyond the melting temperature. Following digestion and ligation, the assembled pAN316a-RNAt-mScarlet was to be transformed into *E. coli* DH5 α to clone, and following extraction, was to be transformed into *S. cerevisiae* BY4741. To assess the control of the RNAt on mScarlet expression at the target temperature of 40°C, the cells would be cultured at the target temperature while measuring mScarlet expression through a plate reader. The cells that fluoresce the most and most quickly would be chosen for the final construct. If the RNAt that was selected was a mutant, the plasmid would be digested and run through gel electrophoresis in order to have the RNAt sequence extracted and sent for Sanger sequencing. This is to determine the identity and structure of the mutant RNAt. The assembly of the final pAN316a-ACC1**-DGA1-OLE1-TPS1-RNAt-BLG2 plasmid (final construct) would take place in a two-step Golden Gate assembly using parts from the Open Yeast Collection (OYC). First, the ACC1**, DGA1, OLE1, TPS1, and RNAt-BLG2 cassettes would be assembled in OYC-dropout-suGFP dropout plasmids. Then, the cassettes would be combined into the final construct in the pAN316a vector. The final construct would be transformed into *S. cerevisiae* BY4741, *S. cerevisiae* POX1 and *S. cerevisiae* PXA1 to compare the lipid profiles of knockout strains. Finally, all transformed strains of *S. cerevisiae* would be assessed in a Nile Red assay to compare and quantify their neutral lipid content.

Designing the RNA thermometer sequence

Synthetic bacterial secondary mRNA structures that were previously created by Neupert et. al (2008) were adopted as the backbone for our design. These simple structures prevent access to important transcriptional regions and block access to the RBS by adding the reverse complement of the conserved Shine Dalgarno (SD) motifs upstream of the region. The goal of this reverse complement region is to form a stem loop structure, where the reverse complement binds to the SD motif to form the stem, and the nucleotides between the two regions will form a loop. Gene expression in eukaryotes and prokaryotes do not tend to use the same machinery or sequences, namely, RNAt are not naturally found in yeast, but are found in bacteria. One of these differences include the use of SD sequences in prokaryotes (*E. coli*), and Kozak Sequences (KS) in eukaryotes, including yeast (*S. cerevisiae*). Both of these motifs function to help ribosomes initiate mRNA translation, with differences in sequence, and positioning relative to the start codon. The RBS is more directly related to translation by providing a site for the ribosomes to bind to, while in the Kozak sequence, it provides a site for the ribosomes to scan the sequences to

find the start codon. We modified the bacterial RNAt region to replace the SD sequence with the eukaryotic specific KS, and corresponding Anti-Kozak Sequence (AKS) upstream. Different KS were also used as different versions were shown to have different effects on gene expression. Rather than creating complex structures that have different expression levels depending on the temperature, our RNAt uses a very simple secondary structure. We designed our RNAt to form a single stem loop that is sufficient to block access to the start codon (by steric hindrance) ribosome initiation site and induce or repress gene expression with a simple temperature shift which will cause the structure to denature and unlock the Kozak sequence and start codon to start efficiently translating the gene into proteins. A pool of newly made constructs synthesized from this method was created and fed into a model to find the most effective design *in silico*.

Constraint-based RNA thermosensor design

We implemented a feedback-driven pipeline that designs *in-silico* RNA thermosensor sequences whose Kozak region stays occluded below a target temperature and becomes exposed above it. Each candidate is modelled as a single strand: a designable flank domain (9-30 nt) followed by their respective Kozak motif. Sequences are produced by NUAD's stochastic local search, which mutates the flank and accepts moves that reduce a weighted sum of constraint violations (Doty, 2023). NUAD is a primarily DNA-based optimization toolbox, as a result of which the strands after each iteration were appropriated to RNA. The thermodynamic mismatch between the RNA using a DNA optimization tool was accounted for by introducing custom penalty scores for the RNAt sequence using the NUPACK toolbox, particularly version 4.0.2.0 (Fornace et al., 2026; Zadeh et al., 2011).

The objective has two parts. A thermodynamic term sweeps NUPACK partition function calculations over target ± 10 °C in 0.5 °C steps and, at each temperature, extracts the unpaired probability of the Kozak bases (Dirks & Pierce, 2003, 2004), combining its mean with the bottleneck (minimum) value (Fornace et al., 2020). Deviations are penalised asymmetrically: exposure below the target is penalised in proportion to how far below it occurs, while incomplete exposure above the target is penalised most heavily just above it. This rewards a sharp switch at the target rather than a gradual melt. Three structural constraints act as filters, holding GC content within 20-80%, hairpin loop size within 4-12 nt, and stem length within 3-25 bp. The loop and stem are measured with an independent maximum-base-pairing dynamic program (Nussinov & Jacobson, 1980), so structural feasibility is not judged by the same model that drives the objective.

The pipeline runs N independent seeded searches to sample diverse local optima, deduplicates the best sequence from each run, and ranks candidates by constraint violation, exposure error, and mean exposure above and below the target. Ranked sequences and the full settings of each simulation are written to disk for reproducibility. The same constraint objects also score

externally supplied sequences across 30-50 °C, allowing designed candidates to be compared against known thermosensors on identical metrics.

Construct Promoter and Terminator Justification

When deciding on the respective promoters and terminators for each Level 1 construct, an important thing to consider was *S. cerevisiae's* ability to do homologous recombination. Having repeats of the same promoter-terminator pair would increase the likelihood of the yeast genome to integrate the plasmid construct part into its chromosome. Another issue was the fact that all the promoters in the iGEM distribution kit had innate 5' UTRs that could interfere with the RNAt. To overcome this, a core promoter sequence was ordered from a paper by Decoene et. al (2019) which removed the innate 5' UTR and was a condensed version of the strong TEF1 promoter. The promoters and terminators for the other genes were chosen based on the expression needed for each gene. For example, the lipid related genes and RNAt needed to be expressed strongly while the other miscellaneous genes could be expressed at a medium-low level to prevent burden on the yeast cell. The promoter expression graph used to select for these parts were taken from experimental data done by Lee et. al (2015) for the YTK and were chosen based on their availability in the 2025 iGEM distribution kit.

dsDNA Synthesis

As double-stranded oligonucleotides are needed for Golden Gate Assembly, dsDNA fragments were synthesized by PCR from single-stranded DNA (ssDNA) templates of the core promoter and three RNAt strands obtained from GenScript. The ssDNA templates and reverse primers were diluted to concentrations of 10µM using nuclease-free water (NFW). PCR reactions were then assembled consisting of 1.25µL of the ssDNA fragments, 12.5µL of 2x Norgen PCR Mastermix, and 10µL of NFW. These were then mixed by pipetting and briefly centrifuged before being placed in a thermocycler. PCR was carried out under the following conditions: initial denaturation at 94-95°C for 2 min, followed by 35 cycles of denaturation at 94-95°C for 15-30 seconds, annealing at 30-40°C for 15-30 seconds, and extension at 65-72°C for 1 min. Annealing temperatures for the RNAt constructs differed for each strand based on the melting temperature predicted on Benchling. A final extension step was performed at 72°C for 5 min before being held at 4-10°C. To verify the PCR products, gel electrophoresis was performed, and the extended DNA was then purified using a gel extraction cleanup, with recovered concentrations of approximately 40ng/µL.

Error-Prone PCR

To introduce random mutations to RNAt, a standard PCR reaction with 51 µL water, 10µL 100mM Tris Cl, 25µL 2M KCl, 4µL each of 5mM or 25mM dNTPs, 1 µL of 5 U/µL Taq DNA Polymerase, 2µL of 100µM forward and reverse primers, 3.5µL 200mM MgCl², 10µL of 200pg/µL template DNA, and an additional 2µL of 25mM MnCl². The thermocycler would be run according to the length of the template DNA, but generally starting at the following settings:

Denaturation at 94°C for 1 minute
Annealing at 60°C for 1 minute
Extension at 72°C for 3 minutes
For 12 cycles (Wilson and Keefe, 2000).

RE Digests

Restriction enzyme digests were used to linearize plasmids and prepare parts for Golden Gate assembly. The 25µL was composed of 500ng-1µg of plasmid DNA, 500ng-1µg of insert DNA, 1µL of each restriction enzyme, 3µL of NEB CutSmart™ buffer, and nuclease free water. The reaction was incubated at 37 °C water bath for 1h followed by heat inactivation at 65°C or 80°C for 20 min, depending on the restriction enzyme. PCR cleanup kits were used to purify the digested product prior to ligation (New England Biolabs, n.d.).

Ligation

Ligation reactions were assembled on ice, combining 2µL of 10x T4 DNA ligase buffer (thawed at room temperature), 50ng of plasmid, 37.5ng of insert, NFW up to a total volume of 20 µL, and 1µL of T4 DNA ligase. Reactions were mixed by pipetting and briefly spun down before being incubated ON at 16°C or, alternatively, at room temperature for 10 min to promote annealing of cohesive ends. The ligase was then heat-inactivated by incubating at 65°C for 10 min. Reactions were then chilled on ice before 1-5µL was transformed into 50µL of competent cells (New England Biolabs, n.d.).

Media Prep

Liquid yeast peptone dextrose (YPD) media was prepared by gently stirring 50g of YPD broth powder with 1L of ddH₂O using a stir bar. To make plates, 20g of agar powder was added to this solution and was stirred before autoclaving for 15 min at 15 psi at 121°C. The final medium was poured into sterile plates in a BSC and cooled until solid (Sagen, 2023).

To make synthetic defined (SD) media, 10g of D-glucose/dextrose was added to a 500mL bottle, and up to 100mL of ddH₂O was added to fill up to the 500mL-indicator. A stir bar was used to gently stir the solution. To a 1L bottle, 0.46g of Yeast Synthetic Drop-out Medium Supplement, 3.35g of Yeast Nitrogen Base without amino acids, and 10g of agar powder were added. Up to 400 µl of ddH₂O was added. Both liquids were autoclaved at 121°C for 15 min, and cooled to 50°C. The D-glucose/dextrose solution was then added to the 1L bottle solution, and sterile water was added to bring the final volume to 1L. The media was poured into sterile plates and cooled for 30min-1hr until solid (Sigma-Aldrich, n.d.).

To prepare lysogeny broth (LB) media, 12.5g of LB powder was added to 450mL of ddH₂O in a 1L bottle. Then, 15g of agar was added to the bottle. Lastly, ddH₂O was added until a final volume of 500mL was reached. Aluminum foil was used to cover the top of the flask before

autoclaving on a liquid cycle (121°C for 15 min). The flask was then cooled to 50°C before being poured into sterile plates (Brydon and Kadimi, 2019).

When supplementing media with ampicillin, a 100mM stock was typically prepared, made of ampicillin powder and filter-sterilized ddH₂O to achieve a final concentration of 100 µM of ampicillin in each plate (Coudert, 2020).

Cell Culturing

To culture *E. coli* DH5α cells for solid media colonies, a small piece of its frozen stock is removed and streaked onto an LB agar plate. The plate is incubated ON at 37°C for 14-18 hours, and stored at 4°C if not being used immediately. For liquid media growth, 5mL of LB broth is transferred into a sterile culture tube. One colony is removed from the solid LB agar plate and inoculated into the culture tube. The tube is then closed and incubated at 37°C, and shaken at 300rpm (Tuttle et al., 2021).

To measure this growth, the incubation time was run in two separate ways. For log-phase growth measurement, the culture was incubated for 2.5-3 hours, where its optical density at 600nm (OD₆₀₀) was measured every 20-30 min. To measure the culture's stationary phase growth, it was grown ON for 12-18 hours. The *E. coli* DH5α cells were harvested mid-log phase by centrifugation (Islam, 2018).

To culture *S. cerevisiae* BY4741 cells, a sterile inoculating loop was used to pick up a yeast colony from glycerol stock and streak onto a YPD agar plate. The plate was inverted and incubated at 25°C or 30°C, where colonies were grown until they reached around 1mm in diameter (approximately 2 days). A starter culture was then prepared by picking a well-isolated colony from the plate, and inoculating it into a culture tube containing 4mL of sterile SD broth. The culture was incubated at 30°C and shaken at 200rpm for 6-8 hours (Tuttle et al., 2021).

To prepare a growth curve of the *S. cerevisiae* BY4741 cells, the starter culture was resuspended and 25mL aliquots of prepared medium for four conditions (control, and three experimental) were each pipetted into separate sterile culture vessels. 25-250µL of starter culture was inoculated at an approximate dilution of 1:100 and 1:1000 into each 25mL media. All four conditioned cultures were incubated at 30°C while being shaken at 200rpm to grow and monitor cell density at OD₆₀₀. Every 2 hours, 1mL of cell culture was removed from each condition. 1mL of uninoculated medium was prepared to use as a blank, and each sample's optical density was measured at 600nm. The cultures were sampled until they reached their mid-exponential (log) phase, defined as an OD₆₀₀ at around 0.4-0.9 (Li et al., 2024).

To harvest the *S. cerevisiae* BY4741 cells, 5-10mL of each experimental culture condition was transferred into a centrifuge tube. Each tube was centrifuged at ~3000-5000 x g for 3-5 min,

forming a cell pellet. Afterwards, the supernatant was discarded. The cell pellet was then resuspended in sterile distilled water or transformation buffer before proceeding with the respective transformation protocol (Tuttle et al., 2021).

Transformation

E. coli DH5 α was transformed using a standard heat shock method, where a stock of chemically competent cells were prepared using calcium chloride and 200 μ L aliquots were stored at -80°C to use for transformations. Chemically competent cells were thawed on ice for 10min, and then combined with 1-5 μ L of 1 pg - 100ng of plasmid DNA, then incubated on ice for 30min. The cells were then heat shocked at 42°C for 45s in a water bath, before incubating on ice a second time for 5 min. The cells were then supplemented with 950 μ L of LB media and incubated at 37°C, 250rpm, for an hour. The cells were then centrifuged, resuspended, and spread on selective agar plates, before finally incubating at 37°C ON for recovery and selection (New England Biolabs, n.d.).

S. cerevisiae BY4741 was transformed using the lithium acetate method for its speed and efficiency and adapted from Gietz and Woods (2002). An ON liquid inoculation of *S. cerevisiae* BY4741 cells in non-selective YPD media were grown to an OD₆₀₀ of around 0.6-1.0, pelleted to extract the cells. The cells were resuspended in a dilute 300 μ L of 0.1M lithium acetate in 5mL YPD solution, and 30 μ L of these cells were combined with 5 μ L of 10mg/mL ss carrier DNA, 40% PEG 3350 with 0.1M lithium acetate, and 0.5 μ g plasmid DNA suspended in 25 μ L sterile water. The ss carrier DNA and PEG 3350 improves the cell's uptake of the plasmid DNA by binding to the cell wall, preventing non-specific loss and helping the lithium acetate make the cell wall more porous to the plasmid DNA. The transformation mix was incubated at room temperature for 30 min, followed by a heat shock at 42°C for 20 min, before centrifuging at 3000rpm for 2 min to harvest the cells. The transformed *S. cerevisiae* BY4741 was then resuspended in 1mL of sterile water and plated on SD agar and incubated at 30°C for two days for recovery and selection.

Passaging cells

Transformed *S. cerevisiae* colonies were passaged into fresh medium for regrowth. A single transformant colony was selected and restreaked onto a fresh selective agar plate before being incubated at 30°C for 2-3 days until there was fresh isolated growth. A starter culture was prepared by inoculating 2-3 mL of selective liquid medium with a portion of the yeast restreak, collected using a sterile loop, and then incubated at 28°C and 250rpm for 24-48 hours. The starter culture was then transferred to a centrifuge tube and centrifuged at 3000 $\times$ g for 5 min. The supernatant was removed, and 9 mL of selective liquid medium was added to resuspend the cells. The culture was then adjusted to OD₆₀₀ = 0.1 and mixed well to ensure even suspension. The culture was then transferred to a culture tube and incubated at 30°C and 250rpm for 24 hours prior to use in later assays (Bond and Tang, 2019).

Golden Gate Assembly

To assemble the Golden Gate reaction, DNA parts, Golden Gate enzyme mix (Bsal-HF-v2), NFW, and T4 DNA ligase buffer were mixed into a tube and later run on a thermocycler. While the components are added at room temperature, the ligase buffer and the Golden Gate enzyme mix were kept on ice. Into each reaction tube, NFW was added first, followed by ligase buffer, DNA inserts, and lastly the golden gate enzyme mix. The reaction mixture was at maximum 20uL. Once all reactions were pipette mixed, they were run on a thermocycler with the following program:

37 °C for 1 min

16 °C for 1 min

The two steps above were repeated for 30 cycles

60 °C for 5 min

After the thermocycler run, samples were stored at -20 °C to be further assembled or transformed (New England Biolabs, n.d.).

For the Golden Gate assemblies, six main linear strands were assembled. Golden Gate constructs all used genes from the Open Yeast Collection kit in the iGem 2024 distribution kit. From the distribution kit, left and right connectors were used, mutagenized coding strands and normal coding strands, and terminators. Single stranded DNA was ordered for promoters and RNAt. BGA1 and DGA1 were ordered through GeneScript. The sequences were constructed as per Figure 1 and 2. Golden Gate assembly was then followed with normal assembly using EcoRI and SpeI restriction enzymes along with the pAN316a plasmid. The linear strands and plasmid were ligated, and purified.

Nile Red Fluorescence Assay

Nile Red, a lipophilic fluorescent dye, selectively stains neutral lipids, allowing lipid content to be assessed through fluorescence intensity. Using a 96 well assay, yeast cells were stained with Nile Red and fluorescence was measured at excitation 485nm and emission 535nm. *S. cerevisiae* BY4741 and *S. cerevisiae* PXA1 cells were prepared according to the Nile Red staining protocol adapted from Rostron and Lawrence (2017), with final OD₅₉₅ readings of 1.030 and 0.988, respectively. The resulting absorbance values at 485nm excitation and 535nm emission are shown in Table1. A 60µg/mL Nile Red in acetone working solution was prepared using 60 µL of 2mg/mL Nile Red stock in acetone to a final volume of 2mL acetone. This stock was re-prepared for every experiment to prevent the evaporation of acetone between experiments from significantly affecting the results of the assay. A 1:1 DMSO:PBS solution was prepared with equal volumes of DMSO and 1x PBS. *S. cerevisiae* cells were harvested by centrifugation for two min at 492 Gs and washed twice with 1x PBS to improve the sensitivity of the assay. The

cells were resuspended in 1x PBS such that the OD₅₉₅ for all samples was around 1.0 when compared to a PBS blank. Using a black, clear-bottomed 96-well plate, each well had 25 µL DMSO:PBS, 25 µL diluted Nile Red solution, and 250 µL of the cell suspension. Each sample was plated as a triplicate and the blank in each row used 250 µL 1x PBS in place of the cell suspension. The plate was read immediately after preparation with wavelength excitation 485nm, wavelength emission 535nm, optimal gain, 10s orbital shake before reading, and reading through the top of the plate (Rostron & Lawrence, 2017).

Gel Electrophoresis

Amplification of RNAi inserts was analyzed using gel electrophoresis. A 1% agarose gel was prepared by mixing 0.5g of agarose powder and 50mL of TAE buffer. The agarose solution was then heated until fully dissolved and allowed to cool to approximately 50°C before 5 µL of Dye-NA-View Stain was added. The resulting dyed agarose solution was then added into a gel caster with an 8-well comb (NUS iGEM, 2023).

PCR Mutagenesis

The coding strand mScarlet3 was transformed into *E. coli* DH5α cells to amplify the DNA and then minipreped to purify. The cds later underwent mutagenesis. On ice, a PCR tube was prepared with 10 uL of PCR mastermix, 2uL of mScarlet, 2.5 uL of forward and reverse primer, and 20uL of NFW. Everything was gently mixed and spun down before putting it inside the thermocycler with the following:

95 °C for 2 min

95 °C for 30 seconds

55 °C for 30 seconds

72 °C for 1 min

The three steps above were repeated for 35 cycles

72 °C for 5 min

4 °C until taken out

The mutagenesis product was then run on an agarose gel using gel electrophoresis (80V for 60 min) and further extracted from its original strand. The extraction was done using a Norgen gel extraction kit (Cat# 13100).

DNA samples were prepared by combining 5 µL of each sample with 1µL of 6x DNA loading dye. The solidified gel was placed in a buffer tank containing 1x TAE buffer and fully submerged. A 1kb DNA ladder (3 µL) and prepared DNA samples (6 µL) were loaded into separate wells. Gel electrophoresis was run at 110V for 25 min. Following electrophoresis, the gel was removed from the apparatus and visualized using the Chemidoc under UV light.

Gel Extraction

DNA fragments were extracted from the agarose gel using the Norgen DNA Gel Extraction protocol (Norgen Biotek Corp, 2026). Following gel electrophoresis, the brightest and most distinct bands corresponding to samples N23 and K23 were selected for extraction. The gel was removed from the Chemidoc for band excision and removal of excess agarose using a spatula, and transferred to microcentrifuge tubes. The N23 and K23 gel slices had masses of 0.19g and 0.21g, respectively.

Binding Buffer G was added to each gel slice at a ratio of three volumes of buffer per volume of gel, resulting in an overall volume of 570 μ L for N23 and 630 μ L for K23. The samples were then incubated at 55°C for 4 min in a water bath to dissolve the agarose. 190 μ L and 210 μ L of isopropanol were added to N23 and K23, respectively, at a ratio of one volume per volume of gel. The dissolved samples were then applied to Norgen spin columns and processed according to the manufacturer's protocol.

Columns were then washed with 500 μ L of Wash Solution A and centrifuged for 1 min to remove residual wash solution. DNA was eluted from each column using 50 μ L of Elution Buffer B.

Results

The pAN316a plasmid was successfully transformed into *S. cerevisiae* BY4741 and validated the URA3 selection system. However, there was no successful growth following assembly with the RNAt components, and the subsequent error-prone PCR was not attempted. There were also issues with the OYC-dropout-suGFP sequence that interfered with transformation, preventing the assembly of the final construct. Additional troubleshooting will be required to validate the proposed construct in the lab, though in-silico results are promising.

From Table 2, it can be clearly seen that *S. cerevisiae* PXA1 emitted greater fluorescence, thus containing more neutral lipid, than *S. cerevisiae* BY4741. Since *S. cerevisiae* BY4741 is a well documented model organism and the relationship between RFU and lipid content is linear (Genicot et al., 2005), the lipid to dry cell weight of *S. cerevisiae* PXA1 can be estimated to be 31.2% higher than BY4741, which is usually around 70mg per gram cell of dry weight when grown in YPD conditions (Kamisaka et al., 2007). From this reasoning, it can be concluded that *S. cerevisiae* PXA1 would have around 91.84mg of neutral lipid per gram of cell dry weight.

Once the RNAt sequences were made, they were run through through the constraint-based software explained in the methods, predicting the melting temperature (defined at the 50% denaturation point) of the candidate sequences in their secondary structure. This helped define

structures that acted optimally, and aided in selection from a pool of RNAs to the top 3 versions to test *in vitro*.

Throughout the RNA design process, several iterations were created guided by the data from the model. Several modifications were made on a control RNA sequence (modified U6) to measure how making changes to the sequence affected the melting temperature. These modifications included adding 1-2 bulges in the stem, adding 5-10 adenines to the loop, shortening the GC clamp, as well as different combinations of the modifications. From the data in Figure 3, we observed that increasing the number of bulges in the stem resulted in a more open RNA conformation, reducing the temperature required to open the structure fully. Increasing the number of adenines in the loop did not make a significant impact on the temperature required to open the RNA, generally opening the structure at the same temperature, but at a slightly faster rate. Shortening the GC clamp had an intermediate impact on the melting temperature, opening up the RNA to a similar extent as adding 1 bulge. By using these findings, several more iterations of RNA sequences were made before arriving at our final 3 sequences.

The first RNA sequence required a temperature of 44.5°C to expose the start codon, while simultaneously 35% of the Kozak sequence was uncovered. At room temperature (25°C), only 1.5% of the start codon was exposed for minimal leaking. A rapid denaturation at 53°C would result in 80% of the start codon to be exposed.

The second RNA sequence was less stringent, with a melting temperature at 39°C that results in 50% of the start codon and 66% of the Kozak sequence to be exposed. At 25°C, 8% of the start codon is uncovered which is more leaky than the first RNA, but still minimal enough. At 52°C, more than 80% of the start codon is exposed to translate the downstream gene.

The third and last RNA sequence was the most liberal with 50% of the start codon exposed at 36.5°C. At 49.5°C, more than 80% of the start codon and 70% of the Kozak sequence is exposed. However, at room temperature, 13.5% of the start codon is also revealed which may cause leaking at non-cooking temperatures.

Discussion

The 5' UTR of the DGA1 was modified to include an RNA sequence to act as a thermosensor to regulate the expression at the translational level. This control works via the formation of secondary structures which block ribosomal access to the RBS. RNA were designed to form a simple stem loop structure at lower temperatures which opens up upon exposure to higher temperatures. We made changes in the sequence which form the stem loop structure blocking the RBS to observe how structural changes affect the stability of secondary RNA structures. The

changes made included adding bulges in the stem (mismatch), adding repeating adenines to make the loop larger, and shortening the GC clamp in the stem.

Introducing a bulge into the stem greatly destabilized the structure at lower temperatures. Considering the stem is the part holding the stem loop structure together, this was expected to have a large impact, where creating gaps in the form of mismatches would make it less stable and susceptible to denaturing. Increasing the number of bulges made this even more apparent with large increases in exposure for each bulge added. The next modification, increasing the size of the loop had subtle effects on the stability. The control, and all of the modified RNAs start rapidly denaturing around 60°C. Upon increasing the size of the loop, the rate of exposure increased, while maintaining the temperature the RNA rapidly started denaturing, suggesting slight denaturing effects. The last modification looked at how the length of the stem affected the stability. Shortening the length of the stem had similar results to the first modification, where there was a large impact on the stability as the stem is the most important part in holding the structure together. By decreasing the length of the stem, there are less base pairs and therefore less hydrogen bonds holding the structure in place. This meant that it would require less energy to open up the structure, making it less stable.

In Figure 3, we observed the impact of combining introducing bulges and increasing loop size. The modifications did not seem to interact with each other, creating additive results, suggesting that modifying the stem or loop independently does not have a major impact on the other. The same trends as before can be seen where increasing the size of the loop increased the rate of exposure, and introducing bulges increased the starting exposure.

One of the main issues we had was transforming the gene constructs within the OYC su-GFP plasmid backbone. A potential solution for this could be to change all the gene inserts and plasmids to the YTK (Yeast Tool Kit) which is widely used and known in industry for yeast genetic engineering. While it was the original plan, due to budgetary constraints, we switched to use the 2025 iGEM distribution kit that we had from years prior. One of the distribution kits was having issues transforming in its entirety, and while iGem was contacted; they advised us it may have been stored incorrectly before reaching us. This coincidentally was the same and only kit containing our OYC su-GFP plasmid backbone. Without the backbone, the Golden Gate assembly could not be completed in the form of a plasmid. The assembly was still done to create a linear golden gate strand, but without the bacterial amplification, following steps involving the digestion, ligation, and purification of this strand resulted in a large amount of product loss. Additionally, when we were ordering the RNA and primers online, they were in a single stranded format to save costs, however this required additional PCR extensions and gel purification which resulted in a lot of product loss. For the future, all the custom designed parts can be ordered as double stranded to avoid further complications.

References

- A Highly Characterized Yeast Toolkit for Modular, Multipart Assembly* | ACS Synthetic Biology | ACS Publications. (n.d.). Retrieved September 29, 2026, from <https://pubs.acs.org/asbcd6/article/4/9/975/1448863/A-Highly-Characterized-Yeast-Toolkit-for-Modular>
- Acetyl-Coenzyme A Carboxylase 1 (ACC1) Assay Kit*. (n.d.). Retrieved September 29, 2026, from <https://bpsbioscience.com/acetyl-coenzyme-a-carboxylase-1-acc1-assay-kit-79315>
- Bond, C. M., & Tang, Y. (2019). Engineering *Saccharomyces cerevisiae* for production of simvastatin. *Metabolic Engineering*, 51, 1–8. <https://doi.org/10.1016/j.ymben.2018.09.005>
- Brydon, W., & Kadimi, S. (2019). *LB Media v1*. <https://doi.org/10.17504/protocols.io.3idgka6>
- Coudert, Y. (2020). *Antibiotic concentrations for LB agar plates v1*. <https://doi.org/10.17504/protocols.io.bghkjt4w>
- Daniel Gietz, R., & Woods, R. A. (2002). Transformation of yeast by lithium acetate/single-stranded carrier DNA/polyethylene glycol method. In *Methods in Enzymology* (Vol. 350, pp. 87–96). Elsevier. [https://doi.org/10.1016/S0076-6879\(02\)50957-5](https://doi.org/10.1016/S0076-6879(02)50957-5)
- Decoene, T., De Maeseneire, S. L., & De Mey, M. (2019). Modulating transcription through development of semi-synthetic yeast core promoters. *PLoS ONE*, 14(11), e0224476. <https://doi.org/10.1371/journal.pone.0224476>
- Dirks, R. M., & Pierce, N. A. (2003). A partition function algorithm for nucleic acid secondary structure including pseudoknots. *Journal of Computational Chemistry*, 24(13), 1664–1677. <https://doi.org/10.1002/jcc.10296>
- Dirks, R. M., & Pierce, N. A. (2004). An algorithm for computing nucleic acid base-pairing probabilities including pseudoknots. *Journal of Computational Chemistry*, 25(10), 1295–1304. <https://doi.org/10.1002/jcc.20057>
- Doty, D. (2023). *nuad: Nucleic acid designer* (Version 0.4.6) [Computer software]. <https://pypi.org/project/nuad/>
- Fakas, S. (2016). Lipid biosynthesis in yeasts: A comparison of the lipid biosynthetic pathway between the model nonoleaginous yeast *Saccharomyces cerevisiae* and the model oleaginous

- yeast *Yarrowia lipolytica*. *Engineering in Life Sciences*, 17(3), 292–302.
<https://doi.org/10.1002/elsc.201600040>
- Ferreira, R., Teixeira, P. G., Siewers, V., & Nielsen, J. (2018). Redirection of lipid flux toward phospholipids in yeast increases fatty acid turnover and secretion. *Proceedings of the National Academy of Sciences*, 115(6), 1262–1267. <https://doi.org/10.1073/pnas.1715282115>
- Fornace, M. E., Huang, J., Newman, C. T., Nanjundiah, A., Porubsky, N. J., Pierce, M. B., & Pierce, N. A. (2026). NUPACK: Computational Nucleic Acid Analysis and Design. *ACS Synthetic Biology*, 15(4), 1426–1441. <https://doi.org/10.1021/acssynbio.5c00817>
- Fornace, M. E., Porubsky, N. J., & Pierce, N. A. (2020). A Unified Dynamic Programming Framework for the Analysis of Interacting Nucleic Acid Strands: Enhanced Models, Scalability, and Speed. *ACS Synthetic Biology*, 9(10), 2665–2678.
<https://doi.org/10.1021/acssynbio.9b00523>
- Genicot, G., Leroy, J. L. M. R., Soom, A. V., & Donnay, I. (2005). The use of a fluorescent dye, Nile red, to evaluate the lipid content of single mammalian oocytes. *Theriogenology*, 63(4), 1181–1194. <https://doi.org/10.1016/j.theriogenology.2004.06.006>
- Gietz, R. D., & Woods, R. A. (2002). Transformation of yeast by lithium acetate/single-stranded carrier DNA/polyethylene glycol method. *Methods in Enzymology*, 350, 87–96.
[https://doi.org/10.1016/s0076-6879\(02\)50957-5](https://doi.org/10.1016/s0076-6879(02)50957-5)
- High Efficiency Transformation Protocol (NEB #C2987H/#C2987I) | NEB.* (n.d.). Retrieved September 29, 2026, from
<https://www.neb.com/en-ca/protocols/high-efficiency-transformation-protocol-c2987>
- iGEM, N. (2023). *Gel Electrophoresis v1*. <https://doi.org/10.17504/protocols.io.n2bvj3dp5lk5/v1>
- Islam, P. (2018). *Optical density measurement for Bacterial solution (Worm Food) v1*.
<https://doi.org/10.17504/protocols.io.vude6s6>
- Kamisaka, Y., Tomita, N., Kimura, K., Kainou, K., & Uemura, H. (2007). *DGAI* (diacylglycerol acyltransferase gene) overexpression and leucine biosynthesis significantly increase lipid accumulation in the $\Delta snf2$ disruptant of *Saccharomyces cerevisiae*. *Biochemical Journal*, 408(1), 61–68. <https://doi.org/10.1042/BJ20070449>
- Li, C., Ouyang, Z., & Liu, J. (2024). Bacterial growth and cultivation. In *Molecular Medical Microbiology* (pp. 155–175). Elsevier. <https://doi.org/10.1016/B978-0-12-818619-0.00070-8>

- Neupert, J., & Bock, R. (2009). Designing and using synthetic RNA thermometers for temperature-controlled gene expression in bacteria. *Nature Protocols*, 4(9), 1262–1273. <https://doi.org/10.1038/nprot.2009.112>
- Neupert, J., Karcher, D., & Bock, R. (2008). Design of simple synthetic RNA thermometers for temperature-controlled gene expression in *Escherichia coli*. *Nucleic Acids Research*, 36(19), e124. <https://doi.org/10.1093/nar/gkn545>
- New England Biolabs. (n.d.-a). *Golden Gate Assembly Protocol for Using NEBridge Golden Gate Assembly Kit (BsaI-HF v2) (NEB #E1601)*. Retrieved <https://www.neb.com/en-ca/protocols/golden-gate-assembly-protocol-for-using-neb-golden-gate-assembly-mix-e1601>
- New England Biolabs. (n.d.). *High Efficiency Transformation Protocol (NEB #C2987H/#C2987I)*. Retrieved <https://www.neb.com/en-ca/protocols/high-efficiency-transformation-protocol-c2987>
- New England Biolabs. (n.d.-b). *Ligation Protocol with T4 DNA Ligase (NEB# M0202)*. Retrieved <https://www.neb.com/en-ca/protocols/dna-ligation-with-t4-dna-ligase-m0202>
- New England Biolabs. (n.d.-c). *Optimizing Restriction Endonuclease Reactions*. Retrieved September 30, 2026, from <https://www.neb.com/en-ca/protocols/optimizing-restriction-endonuclease-reactions>
- Norgen Biotek Corp. (2026). *DNA Gel Extraction Kit: Product Insert* (Nos. PI13100-20). <https://www.google.com/url?q=https://norgenbiotech.com/sites/default/files/2026-06/PI13100-20%2520DNA%2520Gel%2520Extraction%2520Kit%2520Insert.pdf&sa=D&source=docs&ust=1790704097792191&usg=AOvVaw04XuDCV7xvU3k2RxSnrSJK>
- Oelkers, P., Cromley, D., Padamsee, M., Billheimer, J. T., & Sturley, S. L. (2002). The DGA1 Gene Determines a Second Triglyceride Synthetic Pathway in Yeast. *Journal of Biological Chemistry*, 277(11), 8877–8881. <https://doi.org/10.1074/jbc.M111646200>
- PI13100-20 DNA Gel Extraction Kit Insert.pdf. (n.d.). Retrieved September 29, 2026, from <https://norgenbiotech.com/sites/default/files/2026-06/PI13100-20%20DNA%20Gel%20Extraction%20Kit%20Insert.pdf>
- Richards, A. B., Krakowka, S., Dexter, L. B., Schmid, H., Wolterbeek, A. P. M., Waalkens-Berendsen, D. H., Shigoyuki, A., & Kurimoto, M. (2002). Trehalose: A review of

- properties, history of use and human tolerance, and results of multiple safety studies. *Food and Chemical Toxicology*, 40(7), 871–898. [https://doi.org/10.1016/S0278-6915\(02\)00011-X](https://doi.org/10.1016/S0278-6915(02)00011-X)
- Rostron, K. A., & Lawrence, C. L. (2017). Nile Red Staining of Neutral Lipids in Yeast. In C. Pellicciari & M. Biggiogera (Eds.), *Histochemistry of Single Molecules: Methods and Protocols* (pp. 219–229). Springer. https://doi.org/10.1007/978-1-4939-6788-9_16
- Sagen, A. (2023). *Yeast Peptone Dextrose (YPD) medium v1*.
<https://doi.org/10.17504/protocols.io.x54v9d7x1g3e/v1>
- Sharma, P., Mondal, K., Kumar, S., Tamang, S., Najar, I. N., Das, S., & Thakur, N. (2022). RNA thermometers in bacteria: Role in thermoregulation. *Biochimica et Biophysica Acta (BBA) - Gene Regulatory Mechanisms*, 1865(7), 194871. <https://doi.org/10.1016/j.bbagr.2022.194871>
- Sorger, D., & Daum, G. (2002). Synthesis of Triacylglycerols by the Acyl-Coenzyme A:Diacyl-Glycerol Acyltransferase Dga1p in Lipid Particles of the Yeast *Saccharomyces cerevisiae*. *Journal of Bacteriology*, 184(2), 519–524.
<https://doi.org/10.1128/JB.184.2.519-524.2002>
- Tuttle, A. R., Trahan, N. D., & Son, M. S. (2021). Growth and Maintenance of *Escherichia coli* Laboratory Strains. *Current Protocols*, 1(1), e20. <https://doi.org/10.1002/cpz1.20>
- UniProt. (n.d.). UniProt. Retrieved September 29, 2026, from
<https://www.uniprot.org/uniprotkb/P21147/entry>
- Wang, J., Ledesma-Amaro, R., Wei, Y., Ji, B., & Ji, X.-J. (2020). Metabolic engineering for increased lipid accumulation in *Yarrowia lipolytica* – A Review. *Bioresource Technology*, 313, 123707. <https://doi.org/10.1016/j.biortech.2020.123707>
- Wilson, D. S., & Keefe, A. D. (2000). Random Mutagenesis by PCR. *Current Protocols in Molecular Biology*, 51(1). <https://doi.org/10.1002/0471142727.mb0803s51>
- Yeast Drop-Out Bulletin. (n.d.). Retrieved
<https://www.sigmaaldrich.com/CA/en/technical-documents/protocol/microbiological-testing/microbial-culture-media-preparation/yeast-drop-out#products>
- Zadeh, J. N., Steenberg, C. D., Bois, J. S., Wolfe, B. R., Pierce, M. B., Khan, A. R., Dirks, R. M., & Pierce, N. A. (2011). NUPACK: Analysis and design of nucleic acid systems. *Journal of Computational Chemistry*, 32(1), 170–173. <https://doi.org/10.1002/jcc.21596>

Supplementary Information

Table 1. Raw absorption values of the Nile Red stain of *S. cerevisiae* strains BY4741 and Δ PXA1 compared against a PBS blank.

Trial	Blank	BY4741	Δ PXA1
1	15581	36246	42503
2	15200	35279	39660
3	15722	35325	38881
4	15702	35536	45478
5	14953	34660	45328
6	18247	38966	41755

Table 2. Calculated RFU of *S. cerevisiae* strains BY4741 and Δ PXA1 from Table 1 using the formula in Figure 5.

Trial	BY4741	Δ PXA1
1	20345	26602
2	19378	23759
3	19424	22980
4	19635	29577
5	18759	29427
6	23065	25854
Average	20101	26367

Table 3. Genetic sequences of novel parts used in this project. RNAt sequences were developed and tested in silico, and mScarlet primers were designed to be compatible with the RNAt sequences and the pAN316a plasmid.

Part Title	Category	Source	Sequence (5' to 3')
cp_TEF1	Promoter	(Decoene et. al, 2019)	GGTCTCAGGAGAGATTTA AACTCGCGTGTTTTTCTT GTTCTATTACAACCTTTTT

			TACTTCTTGCTCATTAGAA AGAAATACTTGAGACC
RNA _t _1	5' UTR	New	GGTCTCATACTCATTCTT TTTTTTTATCAGTCTGCAG AAAGAAAAAAAAAATGAG AGACC
RNA _t _2	5' UTR	New	GGTCTCATACTCATATTTT TCATCAGTAAAAAAAAA AAAAAAAAAAGGATAAAA AATGAGAGACC
RNA _t _3	5' UTR	New	GGTCTCATACTCATTATTT TCACATGTAAAAAAAAA AAAAAAAAAAGGATAAAA AATGAGAGACC
mScarlet3_fwd	Primer	New	GAACGGTCATGAGTTCGA AATCGAG
mScarlet3_rvs	Primer	New	CTCGATTTCGAACTCATGA CCGTTC

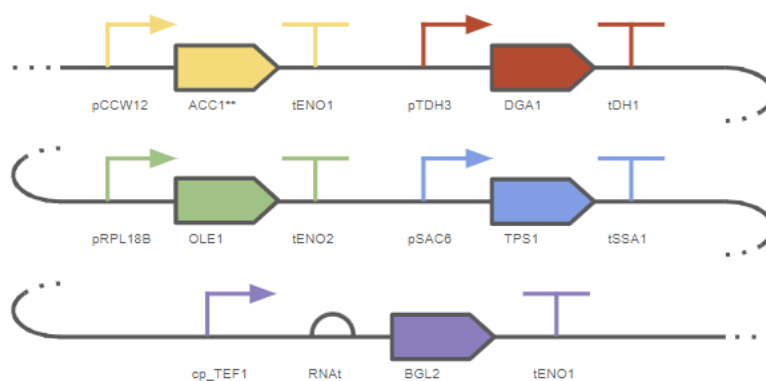


Figure 1. Expected plasmid construct for lipid overexpression and heat-controlled lysis in pAN316A.

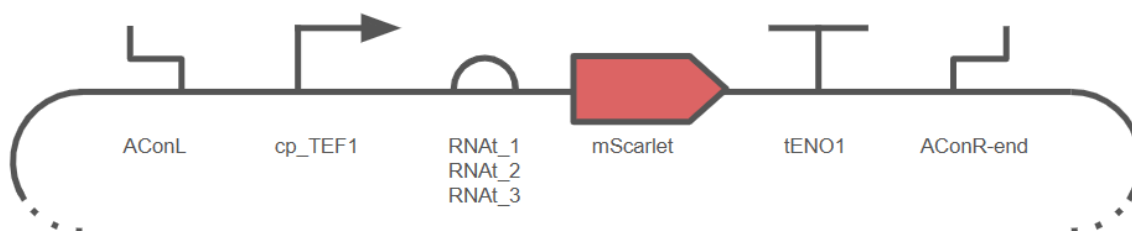


Figure 2. Plasmid construct for gene expression analysis test in RNAi controlled gene in pAN316A.

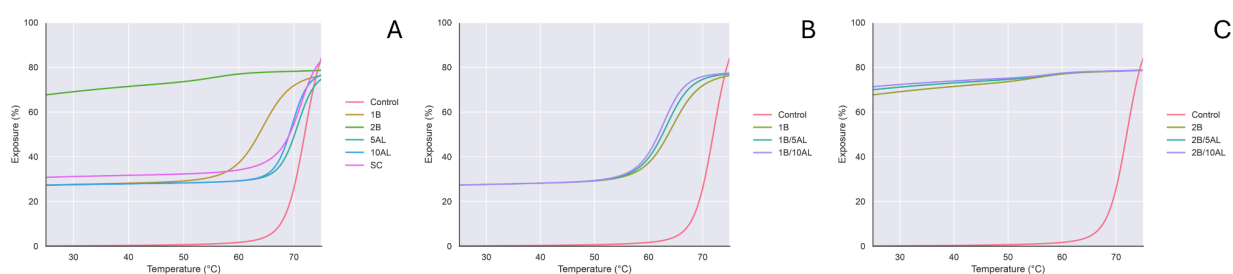


Figure 3. Predicted RNAi exposure based on temperature. **(A)** Basic modifications applied to a standard RNAi sequence. Modifications include addition of 1 or 2 bulges (1/2B) in the stem sequence, 5 or 10 adenines to the loop sequence (5/10AL), or a shortened GC clamp in the stem (SC). **(B)** Predicted RNAi exposure with 1B in combination with 5AL or 10AL. **(C)** Predicted RNAi exposure with 2B in combination with 5AL or 10AL. Data from these modifications and combinations of modifications were used to guide future RNAi sequence design.

RNA Thermometer Optimization Workflow

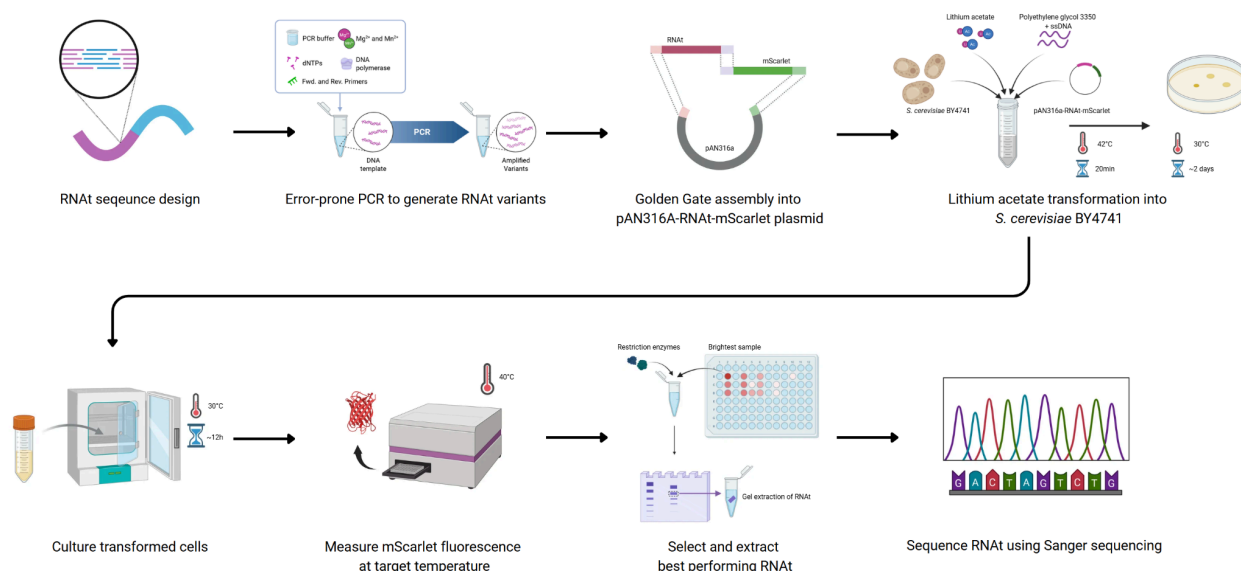


Figure 4. Diagram illustrating the RNA thermometer optimization workflow described in the methods section of the report. Following sequence design, RNAt mutants were to be generated using Error-Prone PCR. The pAN316a-RNAt-mScarlet would be assembled using Golden Gate assembly and would be transformed into *S. cerevisiae* BY4741 using the lithium acetate transformation method. Following recovery, the cells would be cultured at the target temperature while measuring mScarlet expression through a plate reader. The cells that fluoresce the most and most quickly would be chosen for the final construct. If the RNAt that was selected was a mutant, the plasmid would be digested and run through gel electrophoresis in order to have the RNAt sequence extracted and sent for Sanger sequencing. This is to determine the identity and structure of the mutant RNAt.

$$\text{Corrected RFU} = \text{Sample RFU} - \text{Avg. Blank RFU}$$

Figure 5. RFU calculation formula used to generate Table 2 from Table 1. The average blank RFU was calculated to be 15901.